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A comprehensive survey on 2D multi-person pose estimation methods

Chen Wang [a], Feng Zhang [b], Shuzhi Sam Ge [c,*]

[a] School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China
[b] School of Computer Science and Technology, Nanjing University of Posts and Telecommunications, Nanjing, China
[c] Department of Electrical and Computer Engineering, National University of Singapore, Singapore

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ABSTRACT

Human pose estimation is a fundamental yet challenging computer vision task and studied by many researchers around the world in recent years. As a basic task in computer vision, multi-person pose estimation is the core component for many practical applications. This paper extensively reviews recent works on multi-person pose estimation. Specifically, we illustrate and analyze popular methods in detail and compare their pros and cons to fill in the gaps existing in other surveys. In addition, the commonly used datasets, evaluation metrics, and open-source systems are also introduced respectively. Finally, we summarize the development of multi-person pose estimation frameworks and discuss the research trends.

1. Introduction

Human pose estimation, a fundamental yet challenging detection task, aims to obtain body postures from images. Multi-person Pose Estimation (MPE) captures the poses of many persons simultaneously and is one of the most important fields in human pose estimation. It can be applied to many fields such as action recognition (Chéron et al., 2015), person re-identification (Qian et al., 2018), pedestrian tracking (Andriluka et al., 2010), animation production and digital film-making (Siarohin et al., 2019), virtual reality (Pavlakos et al., 2019), human–computer interaction (Weidenbacher et al., 2006), video surveillance (Hattori et al., 2018), self-driving (Murphy-Chutorian et al., 2007), sports motion analysis (Martinez et al., 2017), etc.

The rapid development of deep learning brings about excellent and remarkable achievements in computer vision and has been introduced into various fields, e.g., image classification (Krizhevsky et al., 2012), object detection (Ren et al., 2015; Redmon et al., 2016), gameplay (Mnih et al., 2015), robotic navigation (Mirowski et al., 2018), etc. MPE also benefits from its development greatly. Although there are some existing works (Perez-Sala et al., 2014; Gong et al., 2016; Sarafianos et al., 2016; Presti and Cascia, 2016; Wang et al., 2018; Chen et al., 2020) for human pose estimation, there still lacks a survey to summarize the most recent MPE achievements in detail. Perez-Sala et al. (2014), Aggarwal and Cai (1999), Gavrila (1999), Ji and Liu (2010), Holte et al. (2012) and Moeslund et al. (2006), Moeslund and Granum (2001) introduced early model-based works which model human body as pictorial structures. Chen et al. (2013) and Wang et al. (2018) reviewed works on human motion analysis with depth imagery. Sarafianos et al. (2016) and Presti and Cascia (2016) discussed 3D human action classification using MPE. Chen et al. (2020) presented monocular human pose estimation from the single-person perspective to the multi-person perspective and from 2D pose estimation to 3D pose estimation. However, all these works failed to provide a comprehensive retrospect on MPE. Furthermore, existing surveys do not cover the state-of-the-art methods such as Zhang et al. (2019b), Sun et al. (2018a), Cheng et al. (2020), etc. Our survey gives a clear and holistic knowledge of MPE. We analyze the characteristics of these methods, capture the slight differences between them, and highlight their advantages and motivations. In addition, the commonly used datasets, metrics, and open-source systems are also given.

The remainder of this paper is organized as follows. Section 2 introduces different categories of multi-person pose estimation methods. In this paper, we introduce recent works from two aspects: two-stage MPE and one-stage MPE. In Sections 3 and 4, two-stage MPE methods including top-down and bottom-up approaches are illustrated systematically and comprehensively according to their research motivations such as multi-scale feature learning, data preprocessing, quantization error, etc. Section 5 presents researches on one-stage MPE, which attempt to strike a balance between speed and accuracy. Besides, the commonly used datasets and metrics for MPE are illustrated in Section 6. Meanwhile, several open-source systems are also given in Section 7. Finally, we summarize two-stage and one-stage MPE methods and discuss the future research direction for inspiring new ideas in the MPE field.

2. Categories of multi-person pose estimation

Generally, MPE methods can be divided into three categories according to different classification criteria: (1) deep learning-based vs.

* Corresponding author.
E-mail addresses: wangchen100@163.com (C. Wang), zhangfeng.ac@outlook.com (F. Zhang), samge@uestc.edu.cn (S.S. Ge).

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model-based, (2) two-stage vs. one-stage, (3) graph-free vs. graph matching-based.

Deep Learning-based vs. Model-based: The difference between deep learning-based and model-based methods is whether we define an explicit hand-crafted model to estimate human poses or not. In Fig. 1(a), traditional model-based methods (Andriluka et al., 2009; Fischler and Elschlager, 1973) design a spring model to learn the relations between body parts explicitly. On the contrary, deep learning-based approaches learn the mapping from an image to joint coordinates by explicit models which implicitly capture the relations between body parts from a large number of annotated samples. Although traditional model-based methods have low computation cost, the inconsistent optimization based on hand-crafted features and models leads to unsatisfactory performance. Deep learning-based approaches outperform model-based ones, but need a lot of training data and require high-performance hardware platforms (Toshev and Szegedy, 2014).

Two-stage vs. One-stage: MPE can be generally classified as one-stage methods and two-stage methods according to the number of stages. The two-stage methods can be further divided into top-down and bottom-up methods. Top-down methods mainly involve two steps: person detection and single-person pose estimation (see item (a) in Fig. 2). Specifically, a person detector is used to obtain the bounding box of each person instance and these detected bounding boxes are used to crop images. These cropped images are then fed into a single-person pose estimation method to obtain the pose of each person.

Different from top-down methods, bottom-up methods predict the coordinates of all joints at one time and then use the bipartite graph matching method to assign these coordinates to each person instance (Fig. 2(b)). Although the computational cost of top-down algorithms grows proportionally with the number of detected person instances, they can tackle the scale diversity problem of person instances. The computational complexity of bottom-up algorithms is not related to the number of person instances in the image, but severe occlusions in crowd scenes may hurt the performance of these algorithms.

While the two-stage methods have better accuracy, these methods suffer from inefficient inference which is crucial for practical applications. Furthermore, both the person detector and the single-person pose estimator have a great impact on the overall performance of the two-stage methods. Therefore, researchers resort to combining person detection with pose estimation and propose the one-stage methods. The one-stage methods use a single model to perform body detection and part localization, which is illustrated in Fig. 2(c). A summary of all categories of two-stage and one-stage methods is listed in Table 1).

Graph-free vs. Graph matching-based: According to the assignment mechanism of keypoints, MPE can also be split into graph-free and graph matching-based methods. The graph-free methods are able to assign keypoints to a certain person instance based on the detection result (i.e., the bounding box of a person instance, offset vector or body embedding). Graph-free methods group keypoints in an intuitive way, but their performance relies heavily on the quality of detection results. Unlike graph-free methods, graph matching-based methods treat the keypoint assignment as a dynamic programming problem, but their keypoint assignment is a time-consuming process.

3. Two-stage: top-down methods

Fig. 2(a) shows that the top-down methods typically use a person detector to find person instances in each image and then perform single-person pose estimation for each person instance. In this section, we will introduce the top-down methods from five aspects according to their research motivations: target representation, quantization error, multi-scale feature learning, data preprocessing and non-maximum suppression.

3.1. Target representation

As a supervised learning problem, target representation plays an important role in the performance of MPE methods. Based on the learning target, MPE can be classified into coordinate-based methods and heatmap-based methods (see Fig. 3).

3.1.1. Coordinate-based methods

Thanks to the powerful representation learning capability of Deep Neural Networks (DNNs), Toshev and Szegedy (2014) is the first person who introduced DNNs into the field of human pose estimation and proposed a cascaded framework: DeepPose. The DeepPose adopts a cascade of DNN regressors and iteratively predicts joint coordinates on sub-images by multi-stage refinement.

Direct joint regression (Fig. 3(a)) is intuitive, but the spatial information for each body part is lost. This hurts the accuracy of MPE. Luvizon et al. (2019), Nibali et al. (2018) and Sun et al. (2018b) proposed an end-to-end framework which is described in detail in Section 3.2.

3.1.2. Heatmap-based methods

Different from coordinate-based approaches, heatmap-based methods use the Gaussian heatmap as the learning target to encode the spatial position of joint. Gaussian heatmap was first used in human pose estimation by Tompson et al. (2014) and formulated with an unnormalized two-dimensional Gaussian distribution which centers at the position of joint in Eq. (1).

$$G(x, y, \sigma) = exp(\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2}). \tag{1}$$

where $G(\cdot)$ represents the Gaussian distribution function whose hyper-parameter σ means variance and (x_0, y_0) indicates the accurate joint coordinates. The Gaussian heatmap is generated by using a Gaussian distribution centered at the joint, as shown in Fig. 3(b). When there are more than one person appearing in an image, the learning target can be generated by aggregating several Gaussian distributions with max operation rather than the average operation (Fig. 4). Since the average operation may merge two close peaks into a single one.

Coordinate-based methods output two-dimensional coordinates directly, while heatmap-based methods predict Gaussian heatmaps instead (see Fig. 3). Compared with the coordinates representation, Gaussian heatmap can not only provide spatial information of joints, but also prevent model overfitting which is a common phenomenon in coordinate-based methods. Thus, heatmap-based methods (Tompson et al., 2014; Newell et al., 2016; Chen et al., 2018; Sun et al., 2019) become the mainstream in MPE. However, heatmap-based methods are restricted by the resolution of heatmap which is not consistent with that of input image (see image (b) in Fig. 3). Repeated convolutional operations and pooling operations can reduce the resolution of heatmap, which aggravates the quantization error.

3.2. Quantization error

Gaussian heatmap is an indirect learning target for providing spatial information for pose models. Obtaining the accurate coordinates in the input space from heatmap involves coordinate conversion. Specifically, the coordinates of the maximum activation in the heatmap are integers which are coarse locations for joints. Converting from the coarse coordinates in the heatmap to the input space would aggravate quantization error inevitably. Various methods are introduced to alleviate quantization error in this section from three aspects: prior knowledge-based methods, soft argmax-based methods and offset regression-based methods.

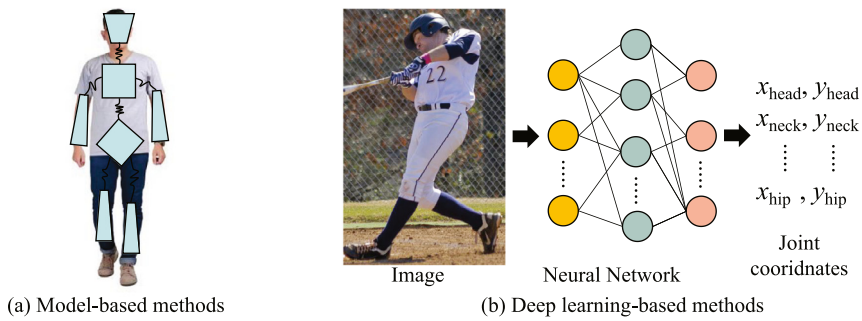

Fig. 1. Illustration of deep learning-based methods and model-based methods. (a): model-based methods simplify human skeleton as pictorial structure where body parts connect with each other by virtual springs. (b): deep learning-based methods map images to joint coordinates by neural networks.

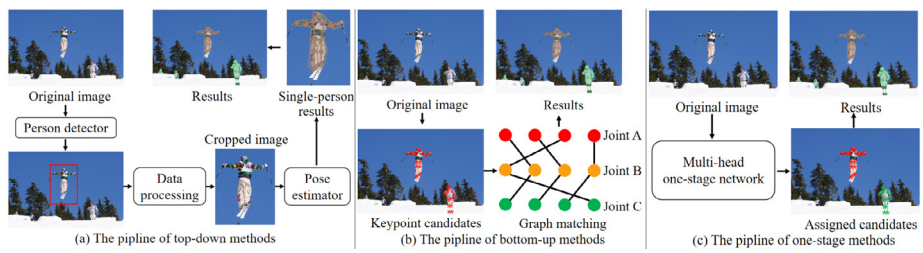

Fig. 2. Pipelines of two-stage and one-stage methods. (a): top-down methods. Persons are detected firstly and person images are cropped after data processing. Then cropped image is fed into single-person pose estimator. (b): bottom-up methods. Keypoints of all persons are predicted at once and assigned to person instances by graph matching. (c): one-stage methods. Original image is fed into multi-head network for keypoint candidates and assignment embeddings simultaneously. According to these embeddings, candidates are grouped as several clusters in different colors. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
The categories of deep learning-based MPE methods.

Categories		Motivations	Subcategories
Two-stage	Top-down	Target representation	(1) Coordinate-based: Toshev and Szegedy (2014), Fan et al. (2015), Luvizon et al. (2019), Nibali et al. (2018), Sun et al. (2018b) (2) Heatmap-based: Tompson et al. (2014), Newell et al. (2016), Chen et al. (2018), Sun et al. (2019)
		Quantization error	(1) Prior knowledge-based methods: Newell et al. (2016), Zhang et al. (2019b), Zhang et al. (2020a) (2) Soft argmax-based: Luvizon et al. (2019), Nibali et al. (2018), Sun et al. (2018b) (3) Offset regression-based: Papandreou et al. (2017), Zhang et al. (2019a), Huang et al. (2019)
		Multi-scale feature learning	(1) Architecture level: Toshev and Szegedy (2014), Wei et al. (2016), Rafi et al. (2016) (2) Module level: Newell et al. (2016), Chen et al. (2018), Sun et al. (2019)
		Data preprocessing	Peng et al. (2018), Toyoda et al. (2019), Fang et al. (2017)
		Non-maximum suppression	Papandreou et al. (2017), Zhang et al. (2020a), Zhang et al. (2019a), Fang et al. (2017)
	Bottom-up	Joint parsing	Pishchulin et al. (2016), Insafutdinov et al. (2016), Cao et al. (2017), Kreiss et al. (2019), Papandreou et al. (2018), Li et al. (2020), Nie et al. (2018)
		Class imbalance	Li et al. (2020), Chen et al. (2018)
One-stage	Bounding box-based	He et al. (2017), Zhou et al. (2019)	
	Embedding-based	Newell et al. (2017), Cheng et al. (2020)	
	Offset-based	Papandreou et al. (2018), Kocabas et al. (2018), Nie et al. (2019)	

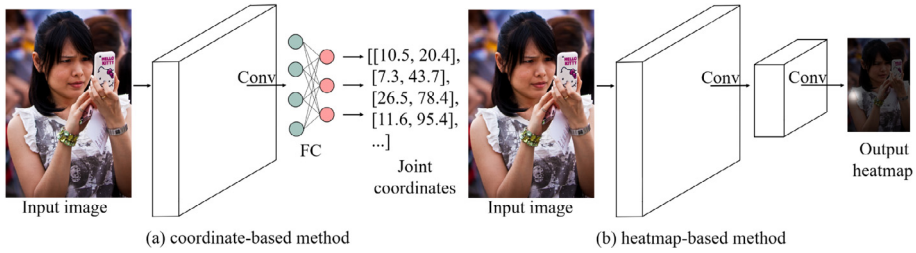

Fig. 3. Illustration of coordinate-based and heatmap-based methods. (a) coordinate-based methods tend to regress joint coordinates directly. (b) heatmap-based methods generate the implicit targets, i.e., Gaussian heatmaps, to provide pixel-wise supervision information and predict heatmaps instead. "Conv" means a convolutional network and "FC" is a fully connected network.

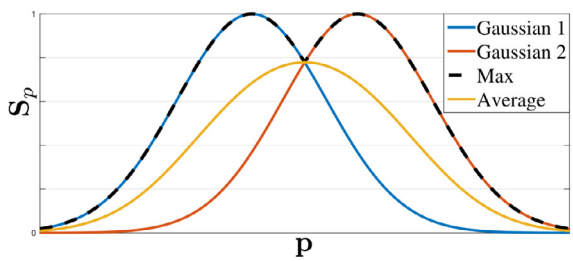

Fig. 4. Illustration of aggregated Gaussian heatmap (Newell et al., 2016). The Max operation keeps the shape of multi-modal Gaussian distribution better, while the average operation keeps only one peak when two Gaussian distributions stay closed.

3.2.1. Prior knowledge-based methods

To get more accurate prediction, Newell et al. (2016) empirically shifted prediction by a quarter of one pixel towards the next highest neighbor in heatmap. Others modeled the heatmap with a prior distribution. Considering the maximum part of output heatmap as the discrete samples of paraboloid, Zhang et al. (2020a) proposed the first-order method to obtain accurate maximal position. Zhang et al. (2019b) revisited the coordinate decoding processing and proposed a novel Distribution-Aware coordinate Representation of Keypoint (DARK) method. DARK approximates Gaussian target heatmap by the Taylor series (Eq. (2)) and gets the peak position of the predicted heatmap according to the first-order and second-order gradient as follows:

$$p(\mu) = p(m) + D'(m)(\mu - m) + \frac{1}{2}(\mu - m)^T D''(m)(\mu - m) \tag{2}$$

where $p(\cdot)$ indicates the predicted heatmap, $D'(m)$ and $D''(m)$ denote the first and second derivative of p evaluated at m respectively.

3.2.2. Soft argmax-based methods

Luvizon et al. (2019), Nibali et al. (2018), Sun et al. (2018b) revisited the post-processing of joint estimation and replaced the argmax operation with the soft argmax module so that the whole framework could be supervised in an end-to-end manner. Specifically, the predicted keypoint coordinate J is aggregated on the standard position field $[x, y]$ weighted with normalized output heatmap $\hat{G}(x, y)$ (see Eq. (3)).

$$J = \sum_{y=0}^{H} \sum_{x=0}^{W} [x, y]^T \cdot \hat{G}(x, y) \tag{3}$$

where H and W represent the height and width of output heatmap respectively.

Sun et al. (2018b) and Luvizon et al. (2019) unified the heatmap representation and joint regression by integral regression, and applied it to 2D and 3D human pose estimation. Nibali et al. (2018) proposed the differentiable spatial to numerical transform module to regress human joints and explored different regularizations to constrain the distribution of predicted heatmap.

3.2.3. Offset regression-based methods

Except soft argmax operation, offset regression is an alternative option to eliminate quantization error. Offset regression-based methods consider accurate joint coordinate as the sum of the integer part and the decimal fraction part and regress both parts respectively. The whole process can be formulated as follows:

$$J = J_{int} + [O_X, O_Y]^T \tag{4}$$

where J_{int} and $[O_X, O_Y]^T$ are the integer part and the decimal fraction part respectively.

G-RMI (Papandreou et al., 2017) outputs $3K$ maps including $2K$ heatmaps and K offset maps, and merges heatmaps with offset maps by Hough voting. Similar to G-RMI, Zhang et al. (2019a) proposed the Offset-Guided Network (OGN), a multi-head network based on ResNet, but obtained offset vector according to the maximal location in heatmap without Hough voting. Huang et al. (2019) adopted the same strategy as OGN to extract the joint coordinates, analyzed the system error resulting from traditional flipping strategy, and proposed unbiased data processing to ensure the alignment between the original and flipped images.

3.3. Multi-scale feature learning

Due to the scale variability of person instances in an image, researchers propose different multi-scale feature learning strategies to mitigate this problem.

These strategies can be classified into two categories: architecture level and module level. The architecture-level methods include DeepPose (Toshev and Szegedy, 2014), Convolutional Pose Machines (CPM) (Wei et al., 2016), Fully Convolutional GoogLeNet (FCGN) (Rafi et al., 2016). Deeppose (Toshev and Szegedy, 2014), the first deep learning-based pose estimation method, adopts a cascade of regressors to estimate joints (Fig. 5). The initial-stage regressor locates keypoints on the full-size images roughly and the consequent regressors taking the image patch as input learn the offset between the prediction from previous stage and the ground truth.

Similar to Deeppose, CPM (Wei et al., 2016) also takes the multi-stage refinement strategy. It adopts a fully differential multi-stage framework where heatmaps from previous stage are always concatenated with low-level features as input to the next stage and the intermediate loss is proposed to prevent vanishing gradients during training. Rafi et al. (2016) built a multi-resolution network containing two FCGNs with half-resolution and full-resolution feature maps as shown in Fig. 6. On one hand, the feature maps from half-resolution FCGN expanded by deconvolutional layers concatenate with features from full-resolution FCGN for heatmaps of different joints. On the other hand, feature maps from layer 13 and layer 17 are concatenated within a single FCGN. Multi-scale feature fusion helps FCGN cover different receptive fields so as to inference joint coordinates with both local and global context.

Another solution to alleviate the scale variability problem is multi-scale feature module. Newell et al. (2016) designed a simple and

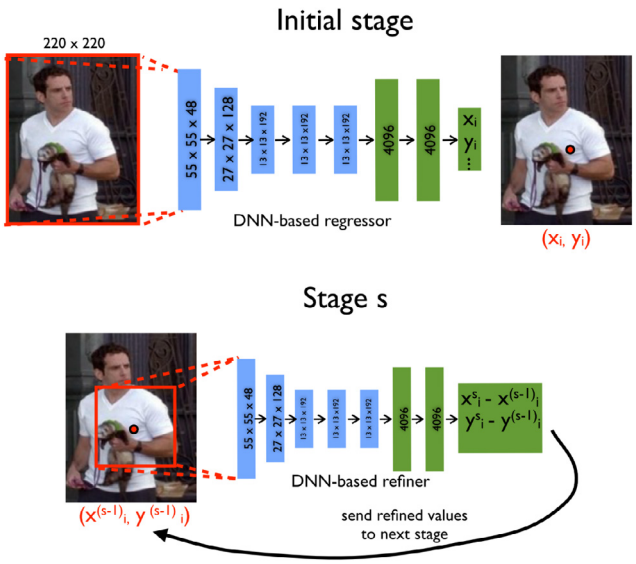

Fig. 5. Illustration of Deeppose framework (Toshev and Szegedy, 2014). DNN-based regressor predicts on the narrowing image patch from previous stage.

symmetric hourglass module (Fig. 7) to capture multi-scale features. Stacked hourglass network consolidates multi-scale features for keypoint prediction by repeating bottom-up, top-down processing across all scales. What is more important is that the symmetric hourglass module can recover the resolution of feature maps to avoid quantization error by nearest neighbor up-sampling. Achieving state-of-the-art results on *COCO 2017 keypoint challenge*, cascaded pyramid network (Chen et al., 2018) adopts GlobalNet for "simple" keypoints and RefineNet for the "hard" samples, as depicted in Fig. 8. GlobalNet leverages feature pyramid networks (Lin et al., 2017) to enhance representation capability. It builds a feature pyramid to enlarge receptive field and fuse features at all levels to form multi-scale representation. Furthermore, all-scale features are fused for the inference of the occluded and invisible joints in RefineNet. Instead of recovering high-resolution representation in previous works (Newell et al., 2016; Rafi et al., 2016; Chen et al., 2018; Lin et al., 2017), Sun et al. (2019) proposed HRNet (Fig. 9) to maintain the high-resolution feature maps and extract multi-scale features.

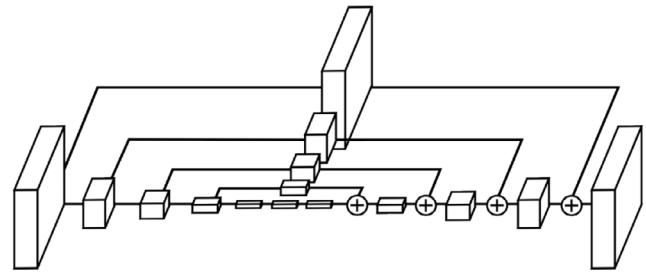

Fig. 7. Illustration of a single "hourglass" module (Newell et al., 2016) where each box corresponds to a residual module and skip layers are used to preserve spatial representations at each resolution.

3.4. Data preprocessing

Data augmentation, e.g. flipping, rotation, scaling, etc., is a basic and effective approach for increasing data diversity, avoiding overfitting and improving the performance of models. Inspired by generative adversarial networks (Goodfellow et al., 2014) and reinforcement learning (Mnih et al., 2015), Peng et al. (2018) designed an adversarial data augmentation network to explore the weakness of Hourglass-based pose detector and achieved better performance by jointly optimizing them together. Considering pose consistency of extreme motions, Toyoda et al. (2019) utilized rotation augmentation to boost the performance of pose model. Fang et al. (2017) adopted pose-guided proposals generator to extend dataset by sampling from Gaussian mixture distribution of the relative offset between the detected and the ground-truth bounding box.

In addition to data augmentation, top-down methods also rely heavily on person detector whose detection results have a great impact on the performance of pose estimator. To reduce the heavy dependence on person detector, Fang et al. (2017) proposed a Regional

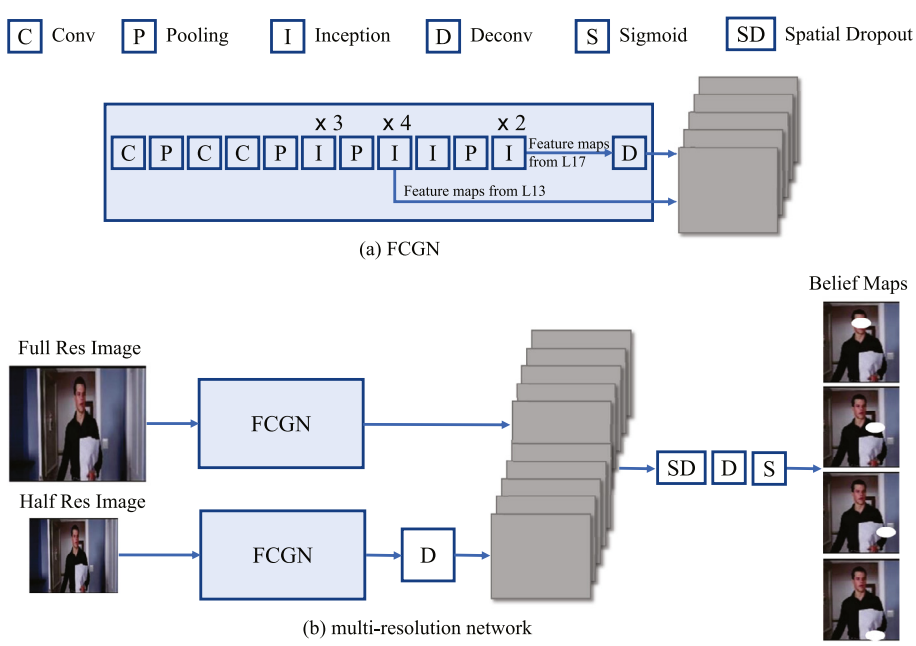

Fig. 6. Illustration of FCGN (a) and the multi-resolution network (b) (Rafi et al., 2016). "Full Res Image" and "Half Res Image" mean a full-resolution image and a half-resolution image respectively. "Belief Maps" are so-called heatmaps.

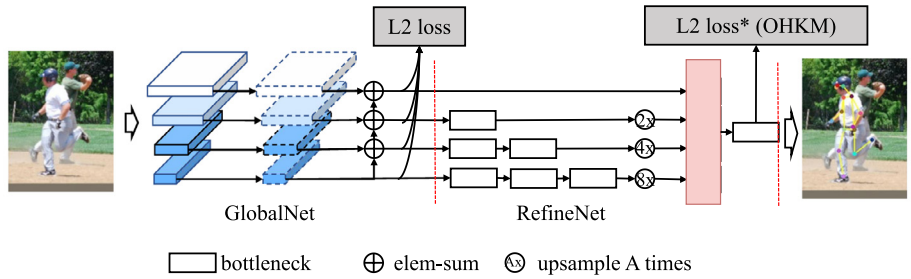

Fig. 8. Illustration of cascaded pyramid network (Chen et al., 2018). Mask RCNN is adopted as the person detector, but not plotted here. "L2 loss*" indicates L2 loss with Online Hard Keypoints Mining (OHKM) which is described in Section 4.2.

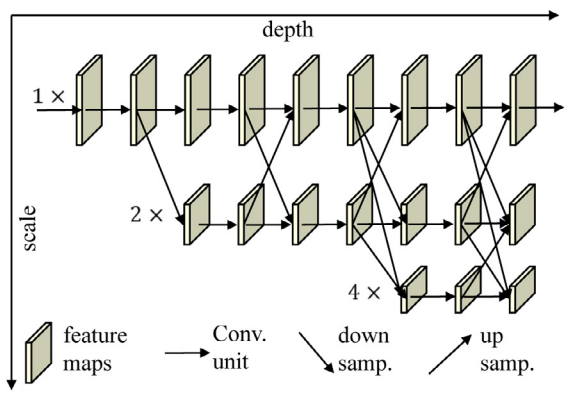

Fig. 9. Illustration of HRNet (Sun et al., 2019) framework. Feature maps at different resolutions are exhibited parallelly with downsample or upsample connection.

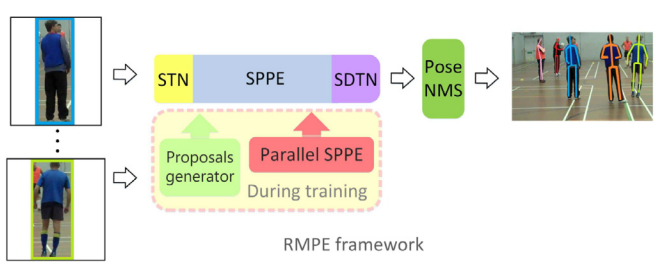

Fig. 10. Illustration of RMPE (Fang et al., 2017) framework, which consists of STN, SPPE, SDTN, pose NMS, proposals generator and parallel SPPE. Proposals generator and parallel SPPE work during training.

Multi-person Pose Estimation (RMPE) framework which includes Spatial Transformer Network (STN) and Spatial De-transformer Network (SDTN) (see Fig. 10). First, original human proposals from the person detector are fed into STN (Jaderberg et al., 2015) to extract high-quality regions of interest. Second, Single-Person Pose Estimator (SPPE) detects joint candidates from these regions of interest. Then, SDTN remaps these candidates back to the original coordinate system. Finally, redundant candidates are removed by pose Non-Maximum Suppression (NMS).

3.5. Non-maximum suppression

Since redundant detection results may exhibit high Intersection over Union (IoU) scores for the person detectors in top-down methods, it is necessary to devise a strategy to remove these redundant detection boxes.

Papandreou et al. (2017) replaced IoU with Object Keypoint Similarity (OKS) in NMS to take keypoints into account. Zhang et al. (2020b) replaced hard-NMS with soft-NMS by keeping and resetting redundant bounding boxes with low scores, which improves 0.3 mean Average Precision (mAP) on COCO dataset. Zhang et al. (2019a) adopted a greedy box generation strategy to produce the best bounding box without NMS. In addition, Fang et al. (2017) employed the parametric Pose NMS (p-Pose NMS) in post-processing to eliminate invalid keypoints from redundant detections.

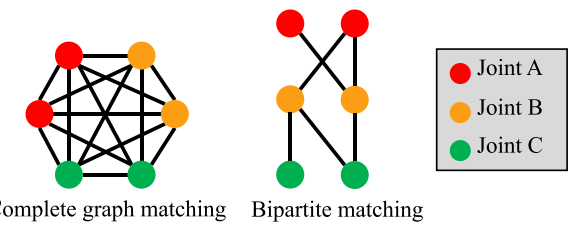

Fig. 11. Comparison between complete group and bipartite matching. *Left*: all nodes connect with each other in complete graph. *Middle*: a node only connects other neighbors with different types. *Right*: legend of joint candidates.

4. Two-stage: bottom-up methods

Unlike top-down methods, bottom-up methods only need one-time network forwarding to regress all keypoints and then assign these joints to different human bodies (see item (b) shown in Fig. 2). This section introduces bottom-up methods from two aspects: joint parsing and example imbalance.

4.1. Joint parsing

DeepCut (Pishchulin et al., 2016) describes a novel partitioning and labeling approach where MPE is cast into the form of an Integer Linear Program (ILP), an NP-hard problem. Although several constraints are imposed for ILP, the complexity of ILP on complete graph is so high that DeeperCut (Insafutdinov et al., 2016) adopts ResNet-based detectors and image-conditioned pairwise terms to reduce running time.

Different from the joint partition on complete graph, Cao et al. (2017) proposed a greedy parsing method based on the tree structure to decrease the search space and complexity of human parsing by bipartite matching in Fig. 11.

OpenPose (Cao et al., 2017) achieved real-time performance and won the first place in *COCO 2016 keypoints challenge*, which mainly consists of body part detection and bipartite matching (Fig. 12). Specifically, it predicts heatmaps and Part Affinity Fields (PAFs) which represent the association of body parts. Association between candidate keypoints is measured by computing the integral over the corresponding PAF. During inference, these keypoint candidates and limbs are viewed as the nodes and edges of a bipartite graph whose edge weights are represented by the association. Thus, the joint grouping task can be reformulated as a bipartite matching problem.

As mentioned in Section 3.2, the heatmap discretization in OpenPose inevitably leads to the quantization error on keypoint location.

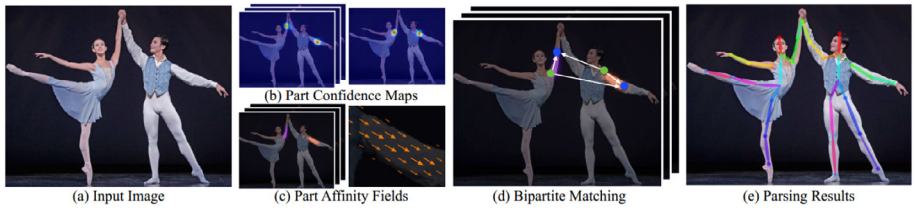

Fig. 12. Pipeline of OpenPose (Cao et al., 2017). OpenPose maps an input image (a) into part confidence maps (b) and part affinity fields (c). Candidates selected from confidence maps serve as nodes in a bipartite graph weighted by part affinity field. After bipartite matching (d), the best parsing results (e) are shown separately.

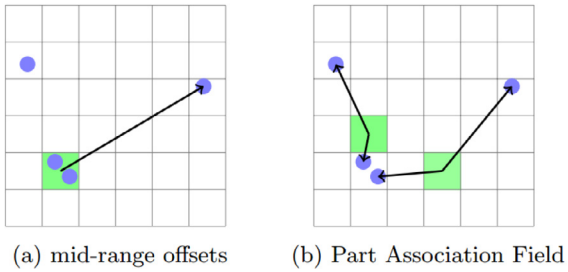

Fig. 13. Illustration of the difference between mid-range offsets (a) from PersonLab and Part Association Fields (b) on a grid of body part map.

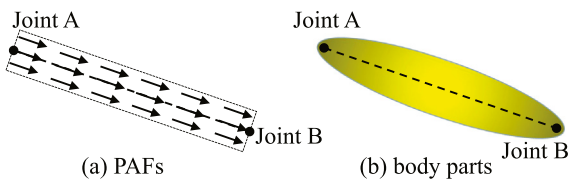

Fig. 14. Illustration of the difference between PAFs (a) from OpenPose (Cao et al., 2017) and "body parts" (b) from Simple Pose (Li et al., 2020).

Therefore, Kreiss et al. (2019) introduced PIFPAF which is composed of Part Intensity Field (PIF) and Part Association Field (PAF).

PIF is designed for precise body part location by applying short-offset vector fields similar to PersonLab (Papandreou et al., 2018) and G-RMI (Papandreou et al., 2017), but high-resolution heatmaps are generated with the unnormalized Gaussian distribution whose standard deviation σ is learned as a part of PIF rather than the bilinear interpolation. In order to learn the body part connection, PAF employs a pair of vectors associated with two body parts but not a mid-range offset in PersonLab (see Fig. 13). PAF is able to tackle joint location of close-by individuals and divide closed joints into distinct instances . Besides, PIFPAF improves its performance on small-scale instances by leveraging the scale-dependent Laplace loss that takes the scale diversity of person instance into consideration.

To reduce model complexity, Li et al. (2020) proposed a novel body association method to replace PAFs in OpenPose with "body parts", namely unnormalized elliptical Gaussian distributions (see Fig. 14). Unlike OpenPose, more limb cues are added for robust prediction.

Considering the high complexity of graph-based parsing methods, Nie et al. (2018) introduced a hourglass-based Pose Partition Network (PPN) which produces joint candidates and partitions simultaneously. The whole parsing process consists of three parts: joint detection, joint partition and local inference. PPN formulates the joint partition as the offset regression where the offset vectors are from joint candidates to the centroid of the person instance. After separating different person instances, joint candidates are selected by greedy local inference.

4.2. Class imbalance

The extreme foreground–background class imbalance is encountered in bottom-up methods such as OpenPose (Cao et al., 2017), PIFPAF (Kreiss et al., 2019), PPN (Nie et al., 2018), etc. To learn hard samples, researchers developed several techniques such as focal loss (Li et al., 2020; Zhou et al., 2019; Lin et al., 2020), Online Hard Keypoints Mining (OHKM) (Chen et al., 2018), etc.

SimplePose (Li et al., 2020) devises the focal L2 loss, a variant focal loss, to balance the gradients of hard and easy samples. The foreground pixels with high confidences and background pixels with low confidences are regarded as easy samples and applied with low weights, while other pixels with middle confidences have high weights in Eq. (5).

$$W(p) = \begin{cases} \|1 - \hat{G}(p) + \alpha\|_2^2, & \text{if } G(p) > thre \\ \|\hat{G}(p) + \beta\|_2^2, & \text{otherwise} \end{cases} \tag{5}$$

where $\hat{G}(\cdot)$ and $G(\cdot)$ are the predicted and ground-truth heatmap respectively. p enumerates all the spatial locations in the heatmap. $thre$, α and β are hyper-parameters for the gradient balance between the foreground and background pixels in heatmap ($\alpha = 0.1, \beta = 0.02, \alpha \gg \beta$).

Similar to CornerNet (Law and Deng, 2018, 2020), Zhou et al. (2019) adopted the focal cross entropy loss to boost the performance of pose model on hard keypoints (see Eq. (6)):

$$L(p) = \begin{cases} -(1 - \hat{G}(p))^{\alpha} log(G(p)), & \text{if } G(p) = 1 \\ -(1 - G(p))^{\beta} \hat{G}(p)^{\alpha} log(1 - G(p)), & \text{otherwise} \end{cases} \tag{6}$$

where $\alpha = 2$ reduces the weights for easy samples and $\beta = 4$ decreases the weights for negative samples further.

Another approach to deal with hard samples is OHKM (Chen et al., 2018). OHKM sorts a batch of training samples by their losses and increases the magnitude of gradients for the top k joints.

In summary, focal loss and OHKM solve the class imbalance problem from different perspectives. Focal loss focuses on hard-to-learn pixels while OHKM emphasizes on hard joints.

5. One-stage methods

One-stage methods, which inherit the strength of two-stage methods and overcome their shortcomings, predict joint candidates and group assignments simultaneously, e.g. associative embedding (Newell et al., 2017), MultiPoseNet (Kocabas et al., 2018), etc.

This section introduces one-stage models based on the joint assignment mechanism and classifies them into bounding box-based, embedding-based and offset-based methods as depicted in Fig. 15. Bounding box-based methods assign joints according to the bounding boxes. Keypoints are assigned to a certain person only when they locate inside the bounding box of this person. Embedding-based methods regress a "tag" for each keypoint, which indicates the identification of a person instance. Offset-based methods predict a vector from joint to person center, which represents the group affiliation.

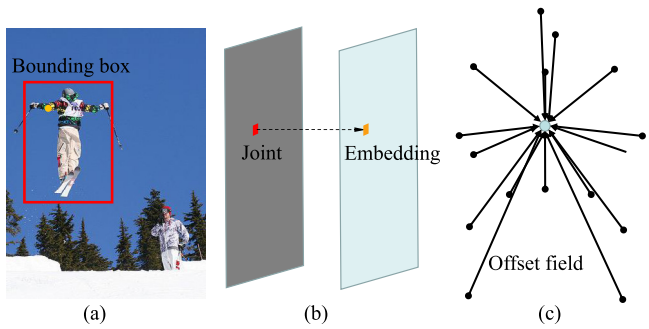

Fig. 15. Illustration of one-stage methods. (a) bounding box-based methods, (b) embedding-based methods and (c) offset-based methods.

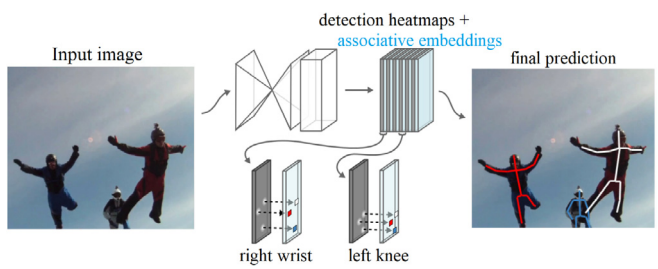

Fig. 16. Illustration of associative embedding (Newell et al., 2017). Hourglass-based network outputs heatmaps and associative embeddings whose colors indicate different persons. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.1. Bounding box-based methods

Mask R-CNN (He et al., 2017) solves MPE problem by predicting bounding boxes and keypoints simultaneously. Keypoints are then assigned to different person instances according to their bounding boxes. Zhou et al. (2019) tackled 2D/3D object detection and pose estimation in a unified framework called CenterNet. The main difference between them is that Mask R-CNN is an anchor-based method while CenterNet is an anchor-free one. Mask R-CNN predicts keypoints in the cropped region of interest extracted by region proposal network, which is similar to the single-person pose estimator. On the contrary, CenterNet directly predicts pixel-wise classification and joint regression on the whole image, and then assembles those joints in the same bounding box.

5.2. Embedding-based methods

Associative embedding (Newell et al., 2017) identifies the group assignment for each joint candidate as a tag regression task in Fig. 16. Newell designed a grouping loss to maximize the distance between tags within a person and minimize the distance between tags across people. HigherHRNet (Cheng et al., 2020) also adopts associative embedding and equips HRNet (Sun et al., 2019) with multi-resolution supervision to solve scale variation challenge.

5.3. Offset-based methods

Papandreou et al. (2018) introduced PersonLab model where short-range offset and heatmap are aggregated via Hough voting for joint candidates like G-RMI (Papandreou et al., 2017), and bidirectional mid-range offset encodes the connection between joints for keypoint assignment. Its recurrent refinement of mid-range offset can effectively decrease the location error. Unlike PersonLab, MultiPoseNet (Kocabas et al., 2018) detected person by a internal detector and fed these proposals into pose residual network to perform keypoint detection, person segmentation and pose estimation. In the Single-stage multi-person Pose Machine (SPM) (Nie et al., 2019), person instance is viewed as a root node and an offset vector from this node to each joint describes its spatial relationship. To tackle the long-range inference, the SPM is extended to a hierarchical one which partitions body parts according to articulated kinematics.

6. Datasets

Dataset plays an important role on MPE. Various datasets are released for pose estimation research such as Leeds Sports Poses (LSP), Frames Labeled In Cinema (FLIC), Max Planck Institute Informatik (MPII), Common Objects in Context (COCO), AI Challenger (AIC), etc. LSP and FLIC belong to single-person datasets, while MPII, COCO and AIC are multi-person datasets. Top-down methods can also utilize existing single-person datasets. In this section, 2D pose estimation datasets and metrics are introduced in detail, and their difference would be compared and emphasized.

6.1. Overview

LSP dataset (Johnson and Everingham, 2010) is released in 2010 and contains 2K pose annotated images of mostly sports people gathered from Flickr. The whole dataset is split into 1K training images and 1K testing images.

FLIC dataset (Sapp and Taskar, 2013) consists of 5,003 images extracted from popular Hollywood movies. In FLIC, almost 20K persons detected by a state-of-the-art person detector are annotated manually in the crowdsourcing marketplace Amazon Mechanical Turk. Besides, FLIC has an extended version which has 20,928 examples.

MPII Human Pose dataset (Andriluka et al., 2014) consists of round 25K images containing over 40K people with annotated body joints. Owing to an established taxonomy of every day human activities, it collects and covers 410 human activities including single person or multi-person inside.

COCO dataset (Lin et al., 2014) is a large and comprehensive dataset for object detection, instance segmentation and keypoint detection. Furthermore, COCO challenge is held every year to promote research in the field of object detection/keypoint detection/instance segmentation. C-OCO dataset includes around 160k images and 100k instances, which are divided into 118K training images, 5K validation images and 40K testing images.

Compared with MPII and COCO datasets, AIC (Wu et al., 2017) aims to provide a large-scale dataset for image understanding. Specifically, AIC consists of three subsets for keypoint detection, zero-shot recognition and Chinese captioning. Here we just introduce the Human skeletal system Keypoint Detection (HKD) which contains 300K high-resolution images with multiple persons and various poses. AIC–HKD dataset is composed of training data (70%, 210K), validation data (10%, 30K), test A data (10%, 30K) and test B data (10%, 30K).

MPII, COCO and AIC–HKD datasets only include a few images in crowd scenes, which do not cover complex environment for practical applications. Recently more and more researchers attempt to handle the occlusion challenge in crowd scene. CrowdPose, the first dataset in crowd scene, is introduced by Li et al. (2019) and contains 10K training images, 2K validation images and 8K testing images.

Table 2 lists all the information of the datasets above and selected example images with annotations are illustrated in Fig. 17.

6.2. Annotation

In LSP, a person instance is labeled with 14 joints, i.e., 0–right ankle, 1—right knee, 2—right hip, 3—left hip, 4—left knee, 5—left ankle, 6—right wrist, 7—right elbow, 8—right shoulder, 9—left shoulder, 10—left elbow, 11—left wrist, 12—neck, 13—head top, as depicted in Fig. 18(a).

Table 2
Multi-person Pose Datasets. Here Jnt. indicates the number of joints and AIC–HKD means the Human skeletal system Keypoint Detection (HKD) of AIC dataset.

Human pose dataset	#Jnt.	Split			#Person instances	Evaluation protocol	Characteristics
		train	val	test-dev			
LSP	14	1k	0k	1k	2k	PCP	Sport activities
FLIC	11	3.987k	0k	1.016k	20k		Movies, upper body
FLIC-full		20k	0k	0k	–		
MPII	16	14.6k	2.7k	6.6k	40k	PCK	Human activities
COCO 2017	17	118.289k	5k	40.671k	100k	OKS	Detection, segmentation, keypoints
CrowdPose	14	10k	2k	8k	80k		Crowd scenes
AIC–HKD		210k	30k	60k	380k		Large-scale images

Fig. 17. Samples from different datasets. *First row*: samples from LSP dataset; *Second row*: samples from FLIC dataset; *Third row*: samples from MPII dataset; *Forth row*: samples from COCO dataset; *Fifth row*: samples from AIC–HKD dataset; *Sixth row*: samples from CrowdPose dataset.

FLIC gives the upper body with only 11 joints, i.e., 0—right wrist, 1—right elbow, 2—right shoulder, 3—left shoulder, 4—left elbow, 5—left wrist, 6—nose, 7—right eye, 8—left eye, 9—left hip and 10—right hip (see Fig. 18(b)).

Fig. 19(a) depicts that each person instance in MPII consists of 16 joints, i.e., 0—right ankle, 1—right knee, 2—right hip, 3—left hip, 4—left knee, 5—left ankle, 6-pelvis, 7-thorax, 8—upper neck, 9—head top, 10—right wrist, 11—right elbow, 12—right shoulder, 13—left shoulder, 14—left elbow and 15—left wrist.

In COCO, each person has 17 annotated keypoints, namely 0—nose, 1—left eyes, 2—right eyes, 3—left ear, 4—right ear, 5—left shoulder, 6—right shoulder, 7—left elbow, 8—right elbow, 9—left wrist, 10—right wrist, 11—left hip, 12—right hip, 13—left knee, 14—right knee, 15—left ankle and 16—right ankle. The facial structure of samples in COCO is more diverse than samples in MPII (see Fig. 19(b)).

In AIC–HKD and CrowdPose datasets, each person is labeled with a bounding box and 14 keypoints: 1—right shoulder, 2—right elbow, 3—right wrist, 4—left shoulder, 5—left elbow, 6—left wrist, 7—right hip, 8—right knee, 9—right ankle, 10—left hip, 11—left knee, 12—left ankle, 13—top of the head, and 14—neck, as shown in Fig. 20. Each keypoint has a flag indicating the visibility.

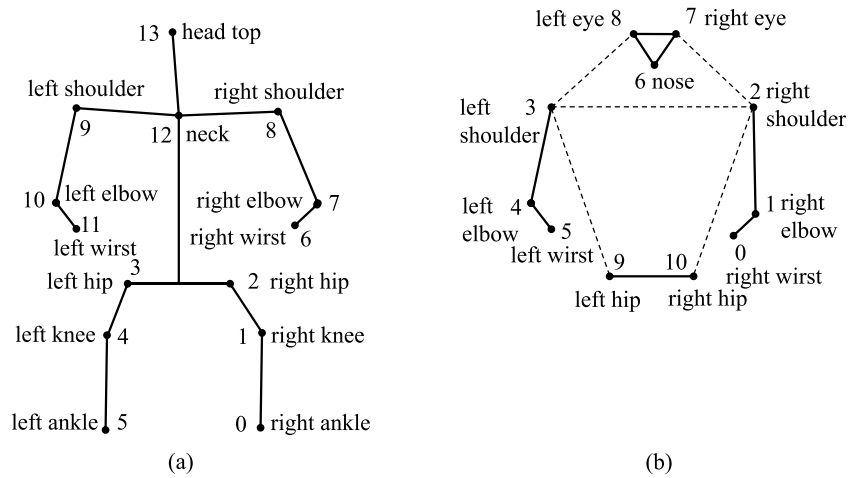

Fig. 18. Annotations of LSP dataset (Johnson and Everingham, 2010) and FLIC dataset (Sapp and Taskar, 2013). (a): in LSP dataset, joints labeled from 0 to 13 are represented with black dots and connected by solid lines. (b): FLIC dataset Only provides the keypoints in upper body. The dotted lines connecting joints is not plotted.

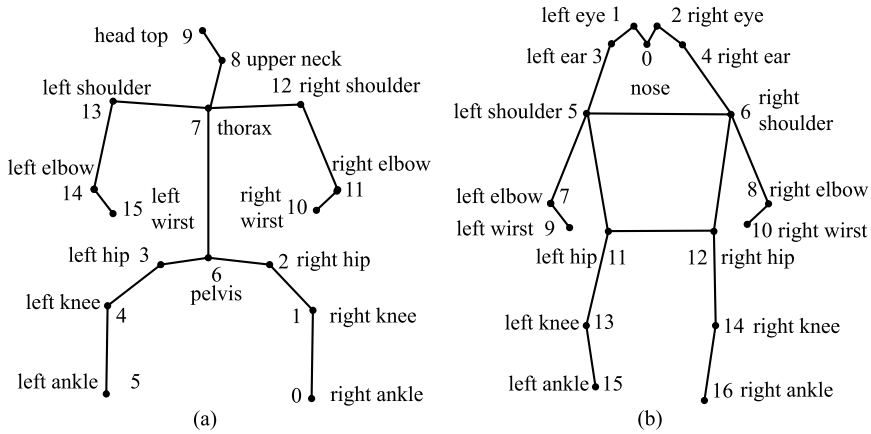

Fig. 19. Annotations of MPII dataset (Andriluka et al., 2014) and COCO dataset (Lin et al., 2014). (a): In addition to the 14 joints in LSP, MPII has two extra joints: pelvis and thorax. (b): COCO has 17 keypoints from 0 to 16. Compared with MPII, it lacks of head top, upper neck, thorax and pelvis, but has some keypoints in head: nose, eyes and ears.

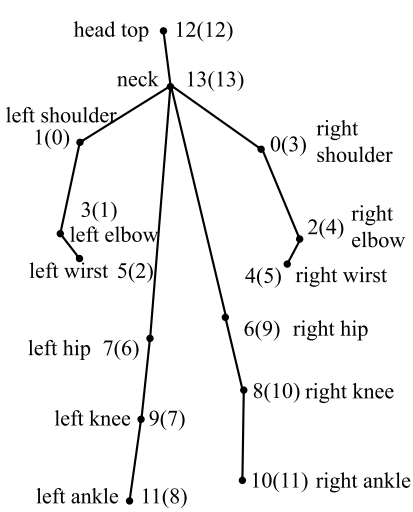

Fig. 20. Annotation of AIC–HKD dataset (Wu et al., 2017) (joint indices inside brackets) and CrowdPose (Li et al., 2019) dataset (joint indices outside brackets). They both have 14 keypoints from 0 to 13. However, the meaning of each index differs from each other. Compared with MPII, AIC–HKD and CrowdPose lack of head top, upper neck, thorax and pelvis, but have additional keypoints in head: nose, eyes and ears.

6.3. Metrics

Percentage of Correct Parts (PCP) is usually used in LSP and FLIC datasets. When the length of a detected limb exceeds a certain threshold, it is viewed as a correct body part. The PCP for a specific part is the percentage of the correct parts in testing set.

In order to measure model performance, MPII adopts the Percentage of Correct Keypoint (PCK) as a evaluation metric (Eq. (7)).

$$PCK = \frac{\sum_i \delta(\frac{d_i}{S} \leq T)}{\sum_i 1} \tag{7}$$

where d_i represents the Euclidean distance for the ith joint between predicted and ground-truth keypoint. The indicator function $\delta(\cdot)$ returns 1 when $\frac{d_i}{S}$ is less than the threshold T. The overall performance metric on the MPII dataset is the mAP where the matching threshold of PCK is set to 0.5.

Different from MPII, COCO uses the OKS to measure the similarity between predicted and ground-truth keypoint (see Eq. (8))

$$OKS = \frac{\sum_i exp(\frac{-d_i^2}{2S^2\sigma_i^2})\delta(v_i > 0)}{\sum_i \delta(v_i > 0)} \tag{8}$$

where $v_i > 0$ indicates the ith joint is labeled. The scale factor S is the square root of the bounding box area, and the variance σ_i indicates the degree of annotation difficulty.

Table 3
Open-source systems. CMU and SJTU indicate Carnegie Mellon University and Shanghai Jiao Tong University respectively.

Open-source System	Affiliation	AP @0.5:0.95	AP @0.5	AP @0.75	AP medium	AP large	Speed(FPS)
OpenPose	CMU	61.8	84.9	67.5	57.1	68.2	10
Detectron(Mask R-CNN)	Facebook	67.0	88.0	73.1	62.2	75.6	5
AlphaPose	SJTU	**73.3**	**89.2**	**79.1**	**69.0**	**78.6**	**20**

A series of Average Precisions (APs) based on OKS, i.e., $AP^{OKS=0.5:0.9}$, $AP^{OKS=0.5}$, $AP^{OKS=0.75}$, AP^{small}, AP^{medium} and AP^{large}, and Average Recalls (ARs), i.e., $AR^{OKS=0.5:0.9}$, $AR^{OKS=0.75}$, $AR^{OKS=0.9}$, AR^{small}, AR^{medium} and AR^{large}, are adopted as the evaluation metrics for COCO, CrowdPose and AIC–HKD datasets.

7. Open-source systems

In order to apply MPE methods to the real-world applications, several open-source MPE systems are released, such as OpenPose, Mask R-CNN, AlphaPose, etc.

OpenPose, the first real-time MPE system, is developed and maintained by Carnegie Mellon University (CMU) which detects various body parts (135 keypoints) efficiently.[1] It supports 2D/3D keypoint detection, camera calibration and single-person tracking in Linux, Windows and Mac OS.

AlphaPose achieves 75.0 mAP on COCO dataset and 82.1 mAP on MPII dataset.[2] It adopts an efficient online pose tracker called Pose Flow (Xiu et al., 2018) to track the identical keypoint from the same person and utilizes spatio-temporal information across frames to reduce the estimation error in videos. Like OpenPose, AlphaPose supports both Linux and Windows.

Mask R-CNN benchmark (Massa and Girshick, 2018) was developed by Facebook in 2017 for object detection, instance segmentation and MPE. Now it is included into Detectron2 (Wu et al., 2019) which only supports Linux.[3]

Table 3 shows that AlphaPose outperforms OpenPose and Mask R-CNN on both accuracy and speed on the COCO test-dev 2015 with a 11G NIVIDIA 1080Ti GPU.

8. Conclusion and future research directions

Multi-person pose estimation is a hot research field of computer vision and attracts researchers from both companies and universities all over the world. This paper reviews recent popular MPE methods from two aspects: two-stage to one-stage. Two-stage methods can be divided into top-down and bottom-up methods. In Section 3, we discuss top-down approaches in different perspectives such as target representation, quantization error, multi-scale feature learning, data preprocessing and non-maximum suppression. Section 4 analyzes bottom-up methods in the terms of joint parsing and class imbalance. Despite top-down methods have advantages on accuracy, bottom-up models with less inference cost are more practical in real-world scenes. In Section 5, we divide one-stage methods into three parts: bounding box-based, embedding-based and offset-based and presents their pros and cons in detail. In addition, this paper gives commonly used datasets and evaluation metrics. Moreover, three popular open-source systems, i.e., OpenPose, Mask R-CNN and AlphaPose, are introduced in Section 7.

In the future, the scale diversity problem could be explored by novel multi-scale representation learning. The severe occlusions occurred in crowd scene might be solved by 3D human body modeling techniques. In addition, the efficiency of pose model plays an important role in practical applications, which can be improved by novel architectures, model compression and pruning, network architecture search, etc.

[1] The source code of OpenPose is available: https://github.com/CMU-Perceptual-Computing-Lab/openpose.

[2] The source code of AlphaPose is available: https://github.com/MVIG-SJTU/AlphaPose.

[3] The source code of Detectron2 is available: https://github.com/facebookresearch/detectron2.

CRediT authorship contribution statement

Chen Wang: Writing - original draft. **Feng Zhang:** Writing - review & editing. **Shuzhi Sam Ge:** Resources, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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